Polynomial Sparsity Testing

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Abstract

contents...

Notations

$\mathbb{F}_q$	Finite field of size q
$\mathbb{F}_q^*$	Multiplicative group of $\mathbb{F}_q$
$\mathbb{Z}_m$	residue class of m
$\mathbb{N}$	Set of natural numbers
$[n]$	The set $\{1, 2, \ldots, n\}$
$\emptyset$	Empty set
$X \sim \mathscr{D}$	Random variable X having distribution $\mathscr{D}$
$\Delta(\cdot, \cdot)$	Relative Hamming distance
$\mathbb{E}[X]$	Expectation of random variable X
$\mathbb{D}[X]$	Variance of random variable X
$\mathbf{w}(l)$	l-th coordinate of $\mathbf{w} \in \mathbb{F}_q^n$
D-evaluation of h over D	Vector of values of h at all points of D
$\mathrm{Unif}(S)$	Uniform distribution over the set S
q	A prime power, unless stated otherwise
$g^{\mathbf{w}}$	$(g^{\mathbf{w}(1)}, \ldots, g^{\mathbf{w}(n)})$
$M_{\mathbf{w},\mathbf{v}}$	$\{g^{\mathbf{w}+\lambda\mathbf{v}} \mid \lambda \in \mathbb{Z}_m\}$
$\mathrm{mon}_{\mathbf{u}}(z_1, \ldots, z_n)$	$\prod_\ell z_\ell^{\mathbf{u}(\ell)}$ in $\mathbb{F}_q[z_1, \ldots, z_n]/(z_1^m - 1, \ldots, z_n^m - 1)$

$\langle \mathbf{u}, \mathbf{v} \rangle$	Inner product of vectors $\mathbf{u}$ and $\mathbf{v}$
$\mathrm{supp}(h(y))$	$\{e \in \mathbb{N}_{\geq 0} : y^e$ has positive coefficient in $h\}$
b-bounded set	Every element of the set ($\subseteq \mathbb{R}$) is less than b
$\mathrm{agr}(e, M)$	Number of agreements of M-evaluation of F and C_m-evaluation of h
$\mathrm{weight}(e, M)$	$\max\{\mathrm{agr}(e, M) : h \in \mathbb{F}[y], h(0) = e, \mathrm{supp}(h) \subseteq S \cup \{0\}\}$
$A \otimes B$	Kronecker product of A and B
$A \odot B$	Hadamard Product of A and B
$\|\mathbf{a}\|$	Hamming weight of $\mathbf{a}$

Contents

CHAPTER 1

SPARSE POLYNOMIAL INTERPOLATION

We shall discuss the paper [BOT88] here.

Definition 1.1 (Sparse Polynomial)**.** We call a polynomial $P \in A[x_1, \ldots, x_n]$ *t-sparse* if it has at most t-many monomials with non-zero coefficient in P.

The main problem of concern for this chapter is the following:

> **Problem 1.2.** How many evaluations are needed to interpolate a t-sparse polynomial P deterministically, if we have the A^n-evaluation of P?

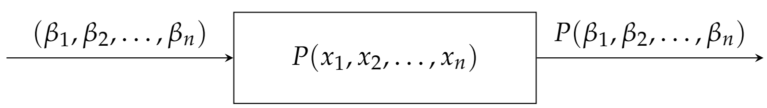

Figure 1.1: Black-box model [BOT88].

Bibliography

[BOT88] Michael Ben-Or and Prasoon Tiwari. A deterministic algorithm for sparse multivariate polynomial interpolation. In *Proceedings of the Twentieth Annual ACM Symposium on Theory of Computing*, STOC '88, page 301–309, New York, NY, USA, 1988. Association for Computing Machinery.

[Bur11] David M. Burton. *Elementary Number Theory*. McGraw-Hill, 7 edition, 2011.

APPENDIX A

APPENDIX

Theorem A.1 (Fundamental Theorem of Arithmetic, [Bur11]). *Every integer $n > 1$ can be expressed as a product of primes. Moreover, this representation is* unique *up to the order of the factors. That is, there exist primes $p_1 \leq p_2 \leq \cdots \leq p_k$ such that*

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdots p_k^{e_k},$$

where each $e_i \in \mathbb{N}$, and this factorization is unique up to reordering of the prime factors.